

# O'RUME DOMINIC URIRIE

AI Engineer · Agentic AI Assurance | LLM and RAG systems, verifiable accountability, measurement instrumentation

## CONTACT

orumedominic@gmail.com  
+44 7402 830944  
rumedominic.com  
linkedin.com/in/dominicrume  
github.com/dominicrume  
Birmingham, UK

## SKILLS

### Languages & Frameworks

Python (production-grade), TypeScript  
/ JavaScript, SQL, FastAPI, Flask,  
Next.js, React

### AI & Agentic Systems

LLM orchestration, RAG pipelines,  
agent tool-use and planning, human-  
in-the-loop control, LangChain,  
LangGraph, CrewAI, local LLMs  
(Ollama), LLM evaluation

### Machine Learning & Research Method

scikit-learn, pandas, PyTorch,  
classification and forecasting, anomaly  
detection, pre-registered experimental  
design, blinded analysis, inter-rater  
reliability (Cohen's  $\kappa$ ), ANOVA

### Cloud, Delivery & Data

Docker, GitHub Actions, GitLab CI,  
Google Cloud Run, Vercel, Fly.io, AWS,  
Supabase, Elastic Stack, OCR  
pipelines, PyPI packaging, automated  
testing

### Verification & Governance

SHA-256 audit chains, on-chain  
anchoring (Polygon), Daml smart  
contracts, STRIDE threat modelling,  
RBAC, secrets management

## CERTIFICATIONS

Microsoft Azure AI Apps & Agents  
Developer  
Certified Blockchain Architect,  
Blockchain Council  
AI Governance & Responsible AI  
Leveraging AI for Governance,  
Regulation and Compliance  
Python, University of Michigan  
Cisco CCNA and CCNP

## PROFILE

AI engineer working where agentic systems meet assurance. I build the instrumentation that answers a question conventional benchmarks do not ask: not whether generated code runs, but whether the agent built the thing it was asked to build. My MSc instrument is open source, pre-registered and published with its own errata; the companion ledger work is catalogued in the Canton Network's official developer hub. Alongside the research I ship production systems: LLM and RAG services, document-verification layers and multi-tenant SaaS, deployed and owned end to end.

## PROFESSIONAL EXPERIENCE

### MCKI Solutions

*AI Engineer, Enterprise Applications*

*November 2025 to present · Birmingham, UK*

- Architect and deploy enterprise AI applications from raw requirements through to cloud-hosted production systems, across student routing, document processing, content automation and housing operations.
- Built an OCR and RAG pipeline at 98% precision that extracts transactions, classifies tax categories and accelerates compliance review.
- Developed an AI student gateway mapping applicant preferences to universities and visa pathways, cutting manual support queries by 40%.
- Own cloud deployment, performance budgets and the CI gates every change must pass, holding a consistent 90+ Lighthouse score.

### VOREM

*AI / ML Engineer and Technical Lead*

*September 2017 to October 2025*

- Moved with the field across two eras: classical ML pipelines and predictive modelling from 2017 (scikit-learn, pandas), then LLM and agentic systems from 2023.
- Engineered a five-model LLM analytics workflow on Google Cloud Run, owning uptime, latency and cost.
- Built a supervised-autonomy quantitative FX and crypto signal system on a nine-module architecture, with API delivery channels and a governance kill-switch.
- Technical project supervisor: led development and testing of AI tools across prompt engineering, generative AI and API integration.
- Monitored production systems using Elastic observability for anomaly detection, and reviewed IT security processes against the findings.

### Simera Global

*AI Advisor (advisory engagement)*

*February 2026 to present · London, UK*

- Advise on the technical design of AI workflows: CrewAI and LangGraph orchestration, verification layers, and secure cross-border data handling for autonomous-agent use cases.

## EDUCATION

### Aston University: MSc Artificial Intelligence with Business Strategy

*2026 · Birmingham, UK*

Dissertation: *Measuring the Unmeasured*, a pre-registered, blinded instrument for auditing the code-quality and governance behaviour of agentic AI coding workflows. Aligned to the Aston and Capgemini Centre of Excellence for Enterprise AI. ML coursework on UK road-accident and cardiovascular data for classification and predictive modelling.

### Covenant University: BEng Electrical & Electronic Engineering

*2014*

Final project: a 5 kVA intelligent changeover inverter. Industrial internship at a 1,020 MW thermal power plant.

## SELECTED PROJECTS

---

### AI Code Quality Auditor: open-source measurement instrument

Python · published on PyPI · [github.com/dominicrume/ai-code-quality-auditor](https://github.com/dominicrume/ai-code-quality-auditor)

An empirical safety harness for agentic AI coding systems, scoring security density, complexity, duplication, specification hallucination and correction frequency across five workflow conditions against fixed specifications, over 600 controlled measurements, with the analysers blinded by interface design rather than by discipline. Published on PyPI across six releases with **3,713 downloads** (pepy.tech, 29 September 2026), shipped as a reusable GitHub Action with a CI gate and a JSON decision record, and served by a live public dashboard.

### Know Your AgenticAI: spend mandates and hash-chained receipts on Canton

Daml · Python · catalogued in the official Canton Developer Hub

Enforces an AI agent's spending mandate inside a Daml choice body (cap, per-period limit, allow-list, expiry, revocation) and seals every attempt into a hash-chained receipt, including the ones the ledger refused, so refusals reach an auditor without exposing the accepted payments behind them. **Three pull requests merged by the Canton Foundation's developer-relations team**, catalogued as a partner tool in the AI Tools category, one of eight AI tools in a sixty-tool catalogue. Published on PyPI as `knowyouragenticai-receipts` with **793 downloads in its first 23 days** and **zero dependencies**, plus a client-side verifier that checks a receipt chain in the browser with nothing uploaded and no network needed.

### Tenant Hub: enterprise housing-operations SaaS

Next.js · Supabase · MCKI Solutions · live April 2026

Multi-tenant platform serving 100+ tenants across housing brands, with role-based access control, brand-level data separation, SHA-256 audit-chain records and CI-enforced deployment controls. The automation saves the operator roughly 100 hours per week and about £80,000 per year.

### VeritaPort: evidentiary audit layer for a legal-tech platform

FastAPI · Polygon · HMAC-SHA256 · client engagement, integrated by a London disputes practice

Built the cryptographic audit-trail layer for an institutional decision-capture platform: live Polygon audit stamping with document tamper verification, so any third party can verify record integrity in seconds without per-user wallets. Followed it with the enterprise API layer, adding scoped API keys, per-firm multi-tenant isolation, HMAC-SHA256 signed webhooks and an OpenAPI specification. Chose 404 over 403 on cross-tenant reads so a firm cannot learn that another firm's matter exists. Both pull requests were reviewed and merged by the client.

### AMIAF Seal: on-chain provenance for physical artwork

Solidity · EVM · Polygon · Next.js · live at a festival in Abuja, 30 September 2026

Gives a physical artwork a tamper-evident record anchored to a public chain: a fingerprint derived from the artist's own photograph, a public certificate, and a hash chain anyone can re-verify afterwards. Ran nine hours unattended on the day. **94 works sealed, 94 of 94 previous-hash matches, zero broken links, zero duplicate fingerprints, and one Polygon transaction anchoring the whole day.** 106 works on the public record since. Counted from the raw backups rather than from the dashboard.

## RESEARCH, IP & RECOGNITION

---

- **UK patent application GB2611754.9**, filed with the UK IPO on 20 May 2026. Pending and not yet published; filing receipt available on request. Covers the KYA framework for verifiable AI accountability and provenance.
- **Named technical partner on an Innovate UK Frontier AI consortium bid** with Learncast and the University of Birmingham, with the patent-pending KYA framework as background IP. Ownership and terms confirmed in writing by the commercial lead and by the University's Impact and Engagement Manager, School of Computer Science.
- **Third place, CANTOR8 "Build on Canton" hackathon**, London, competing as Vorem. Awarded from a \$25,000 prize and development pool, judged by panels drawn from across the Canton ecosystem on technical, security, institutional and commercial criteria.
- **ACM Europe Seasonal School on Responsible AI**, Aston University, September 2026. Selected through a scored international application round that invited only the top 30–50 candidates. My team was awarded second prize in the school's project competition.
- **Unsolicited security review of OpenZeppelin's Canton contracts.** Filed three issues showing that authorisation fences ship without tests that fail when the fence is removed: 25 of 30 in `canton-stablecoin` including the oracle price guard, 24 of 32 in `canton-token-template`, and three access-control fences in `canton-contracts`.
- **FrontierTechX**, University of Birmingham, March 2026. Panellist, "Agentic AI: From Copilots to Autonomous Enterprises". Also hosted and spoke at two further Birmingham technology events in 2026.
- **MSc dissertation** validated by two independent raters (Cohen's  $\kappa = 0.870$  rater-to-rater), with the erratum that overturned one of its own cleanest findings published rather than quietly corrected.
- **Author of five books** on AI and blockchain, available through Waterstones and Amazon.

## REFERENCES

---

Contact details for both are available on request.

### MCKI Solutions

Employer, November 2025 to present

### VOREM

Employer, 2017 to 2025